

Laboratory Work 5: Deep Learning Clustering

Sekenova Balym. ID: 23B031433

1 Introduction

In this lab we study deep learning clustering and compare unsupervised and supervised approaches. The main idea is to understand how models work when they do not have labels, and how this is different from models that are trained with correct answers.

We use the MNIST dataset, which contains images of handwritten digits from 0 to 9. In the first part, we apply an autoencoder and then use clustering to group similar images. In the second part, we use a convolutional neural network (CNN) to classify digits using labels.

The goal of this lab is to see the difference in performance between these two approaches and understand why supervised learning usually gives better results for classification tasks.

2 Methodology

In this section we describe the methods used in the lab. We apply an unsupervised approach based on an autoencoder and clustering, and a supervised approach using a convolutional neural network (CNN).

2.1 Unsupervised Clustering

In the first part of the lab we use an autoencoder to learn a compressed representation of the MNIST images. Each image of size 28x28 is converted into a vector and then reduced to a latent space of size 16. The model is trained to reconstruct the input images, so it learns general features like shape and structure.

After training, we extract the latent vectors and apply K-Means clustering with $k=10$. The idea is that similar digits should be grouped into the same cluster, even though the model does not use labels during training.

To visualize the latent space, we first use PCA, but the results are not very clear because PCA is a linear method. Then we apply t-SNE, which shows the structure better and makes clusters more visible.

Finally, we compare clusters with true labels using a confusion matrix. This helps to understand how well the clustering corresponds to actual digit classes, even though labels were not used during training.

2.2 Supervised CNN

In the second part of the lab we use a convolutional neural network (CNN) for image classification. Unlike the previous approach, this model is trained using labels, so it learns to predict the correct digit directly.

The CNN consists of convolutional layers that extract important features from images, followed by fully connected layers for classification. The model is trained using cross-entropy loss, which helps to improve prediction accuracy.

After training, we evaluate the model on the test dataset. The results show very high accuracy, close to 99%, and the confusion matrix is almost perfectly diagonal. This means that the model correctly classifies most of the digits.

This part demonstrates how supervised learning can learn more discriminative features compared to unsupervised methods.

3 Results

The autoencoder training loss decreases steadily during training, which shows that the model learns to reconstruct the input images correctly. However, the clustering results are not very well separated. The PCA visualization (Figure 1) shows overlapping clusters, while t-SNE (Figure 2) provides a clearer view of the latent space, where some grouping of digits can be observed.

The confusion matrix for clustering (Figure 3) shows that many digits are mixed within the same clusters, which means that clustering does not align well with true labels.

In contrast, the CNN model achieves very high accuracy of about 99%. The confusion matrix (Figure 4) is almost perfectly diagonal, indicating that most digits are classified correctly.

Table 1 summarizes the accuracy results, clearly showing the performance difference between clustering and CNN.

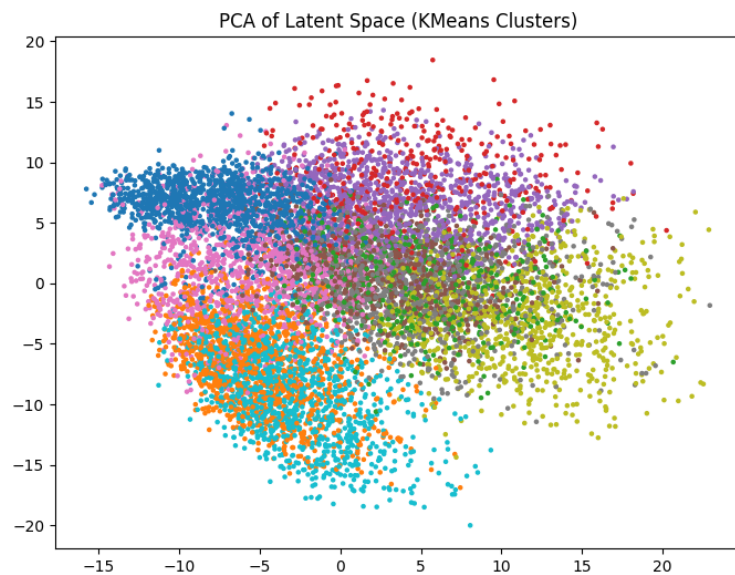


Figure 1: PCA visualization of latent space

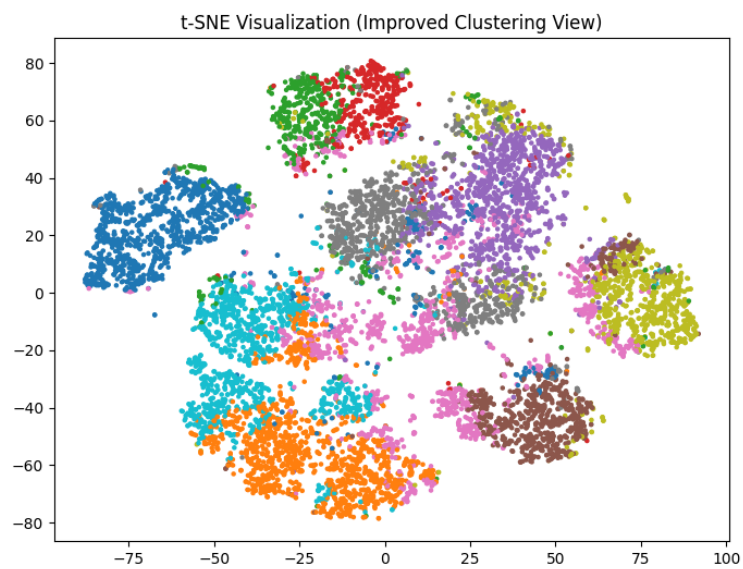


Figure 2: t-SNE visualization of latent space

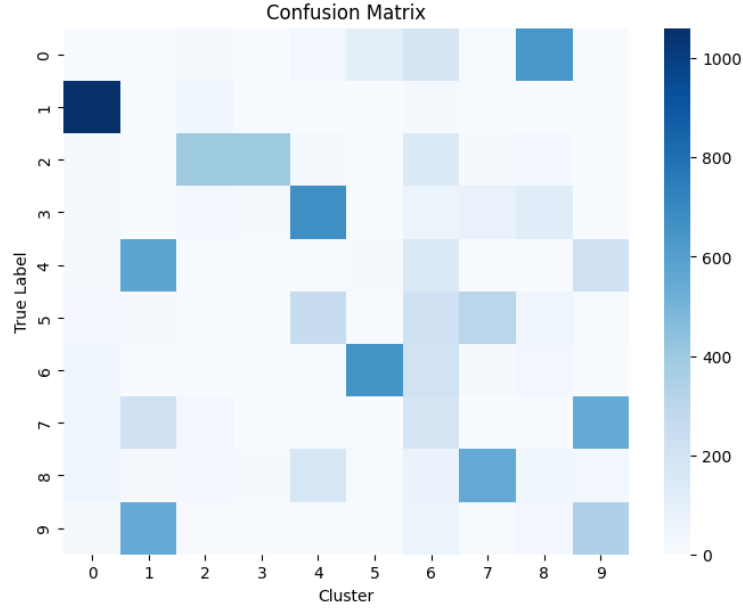


Figure 3: Confusion matrix for clustering

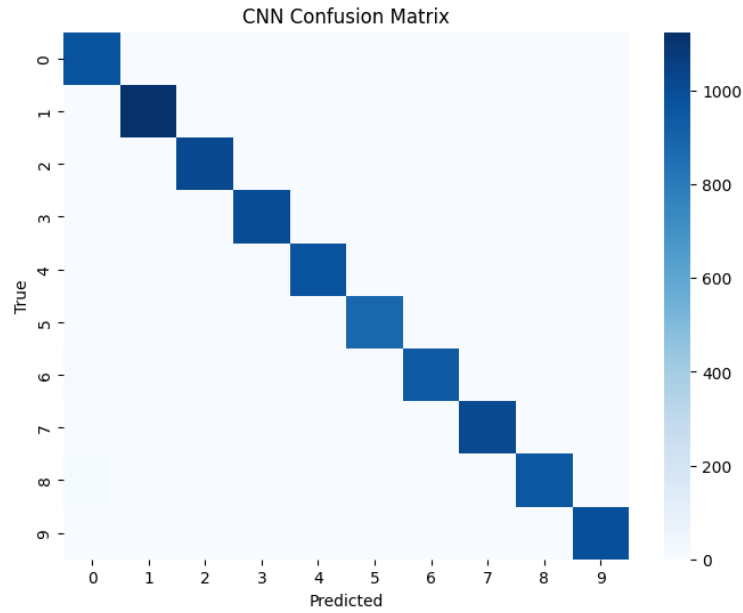


Figure 4: Confusion matrix for CNN

Model	Accuracy
Clustering (Autoencoder + KMeans)	0.637
CNN	0.990

Table 1: Comparison of model performance

The clustering approach achieves an accuracy of approximately 63.7%, while the CNN model reaches around 99%, clearly showing the difference in performance.

4 Discussion

The results show a clear difference between unsupervised clustering and supervised learning. The main reason why the CNN performs better is because it uses labels during training, while clustering does not.

In the autoencoder approach, the model is trained only to reconstruct the input images. Because of this, it learns general features such as shapes and strokes, but it does not learn the actual meaning of digits. As a result, digits that look similar, for example 3 and 5 or 4 and 9, are often grouped into the same cluster, even if they belong to different classes.

Another reason is that K-Means assumes that clusters are simple and well-separated, but the MNIST dataset has more complex structure. This leads to overlapping clusters and a messy confusion matrix.

On the other hand, the CNN is trained directly with labeled data, so it learns discriminative features that help to clearly separate different digit classes. This results in very high accuracy and a nearly perfect confusion matrix.

Overall, this shows that supervised learning is more suitable for classification tasks, while unsupervised methods are more useful for exploring patterns in data.

5 Conclusion

In this lab we explored both unsupervised and supervised approaches for working with the MNIST dataset. The autoencoder with K-Means clustering was able to capture some structure in the data, but the results were not very accurate for separating digit classes.

In contrast, the CNN model achieved much better performance, with high accuracy and clear classification results. This is because it was trained using labeled data and could learn more specific features for each digit.

This lab shows that unsupervised learning can be useful for understanding data and finding patterns, but supervised methods are more effective when the goal is accurate classification.